Bluetooth RC car

Components

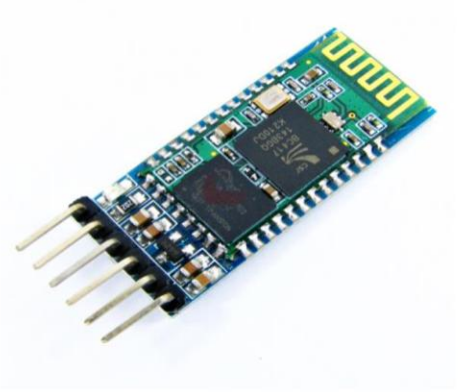
- HC-05 Bluetooth transceiver module
- L298N motor driver module
- DC motors and wheels x2
- Batteris ~12 Volt



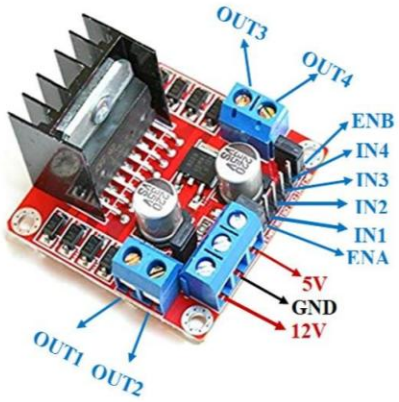
Development Board (AT89C51)



Lithium batteries



HC-05

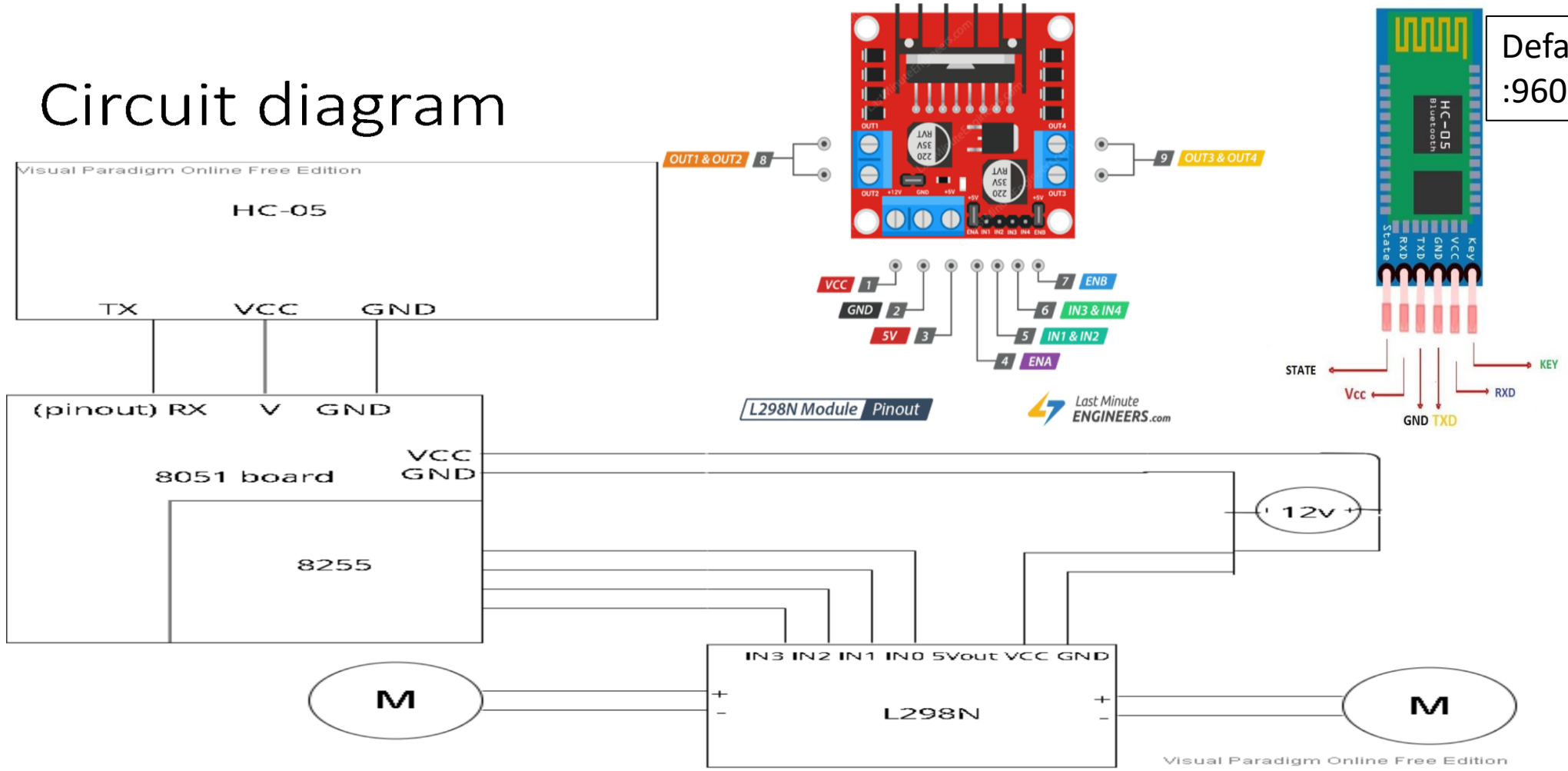


L298N



Wheels

Circuit diagram



Default baud rate :9600

Init serial port

```
    ;init serial port
mov scon, #50h ;SERIAL PORT MODE == 1, 8-bit UART,
               ;SET RI WHEN RECEIVE STOP BIT.
mov tmoD, #20h ;TIMER 1 MODE == 2,
               ;C/T == 0, INTERNAL CLOCK SOURCE,
               ;GATE == 0, SOFTWARE-CONTROLLED TIMER;
mov th1, #250 ;BAUD RATE 9600
setb tr1 ;TURN ON TIMER 1
```

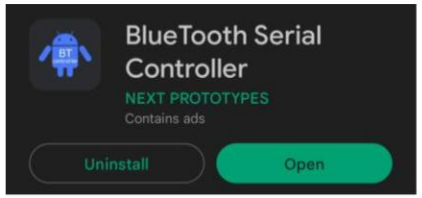
Init 8255 ppi (l298n, leds, …)

```
    ;init 82C55
    ;(port A output,port B input)
mov A, #0x82 ;CONTROL WORDS
mov DPTR, #0xA003 ;HARDWARE-DEFINED, LEDS
movx    @DPTR, A
mov A, #0x80 ;CONTROL WORDS
mov DPTR, #0x9003 ;HARDWARE-DEFINED, L298N
movx    @DPTR, A
```

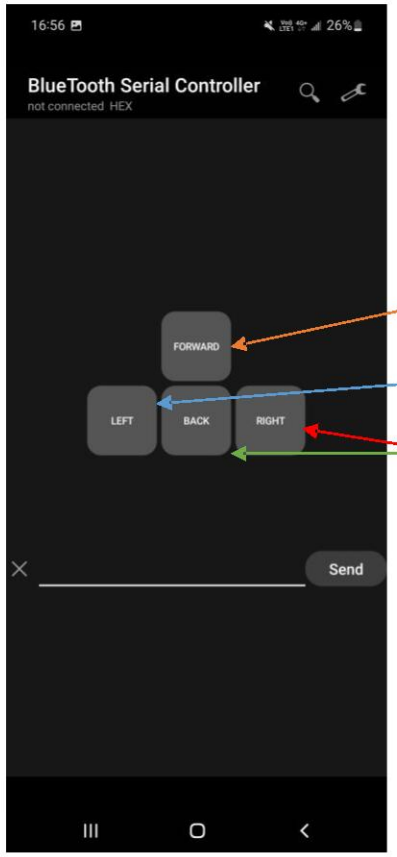
Smartphone as controller

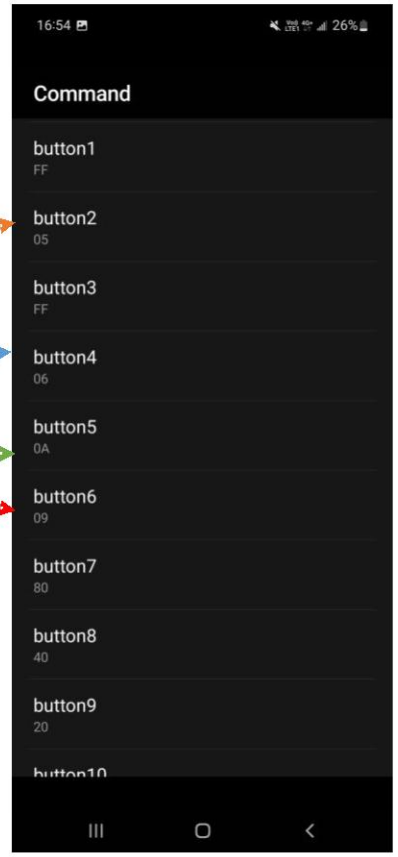
L298N Motor Controller Truth Table			
ENA	**IN1**	**IN2**	**Description**
0	N/A	N/A	Motor A is off
1	0	0	Motor A is stopped (brakes)
1	0	1	Motor A is on and turning backwards
1	1	0	Motor A is on and turning forwards
1	1	1	Motor A is stopped (brakes)

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Google Play Store



IN1, IN2: MOTOR A
IN3, IN4: MOTOR B





00000101 FORWARD

00000110 BACKWARD

00001010 ...

00001001 ...

Function 1 : Change speed (1/2)

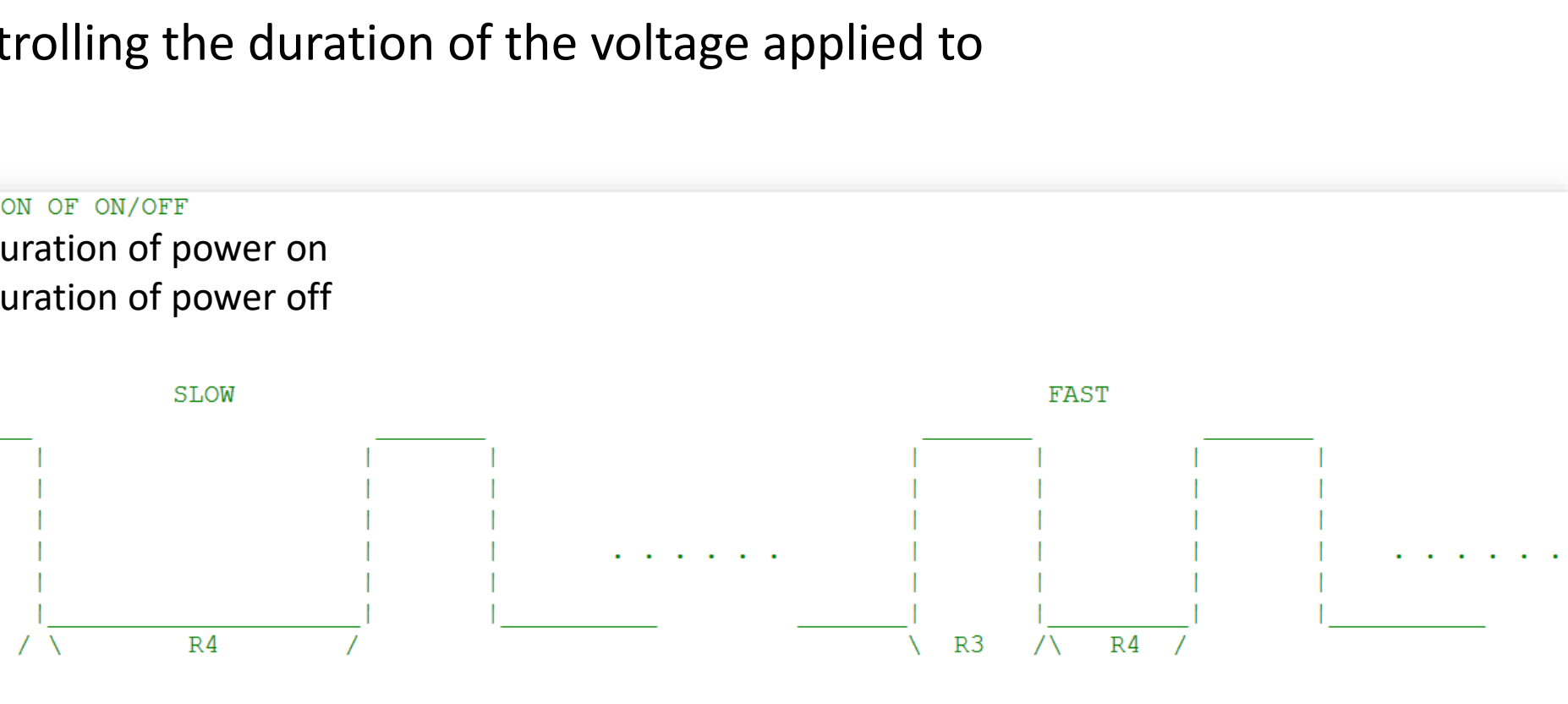
change the speed by controlling the duration of the voltage applied to the motors.

```
SPEED: ;CHANGE SPEED BY CHANGING DURAION OF ON/OFF
       ;R3 == DURATION OF ON
       ;R4 == DURATION OF OFF
    jnb acc.0, SLOW
    ljmp FAST

    SLOW:          ;
        mov R3, #1;
        mov R4, #3;
        ljmp HERE ;
                  ;
    FAST:          ;
        mov R3, #1;
        mov R4, #1;
                  ;
    HERE:
        ljmp MAIN
```

R3 : duration of power on

R4 : duration of power off







Function 1 : Change speed (1/2)

Turn on motors,
call delay subroutine,
turn off motors,
call delay subroutine.

```
PASS: ;ALTERNATING POWER ON AND OFF
    mov DPTR, #0x9000
    movx @DPTR, A        ;
    mov A, R3            ;
    acall _DELAY         ;
                         ;
        ;BRAKE           ;
    mov A, #00000000b    ;
    movx @DPTR, A        ;
    mov A, R4            ;
    acall _Delay         ;

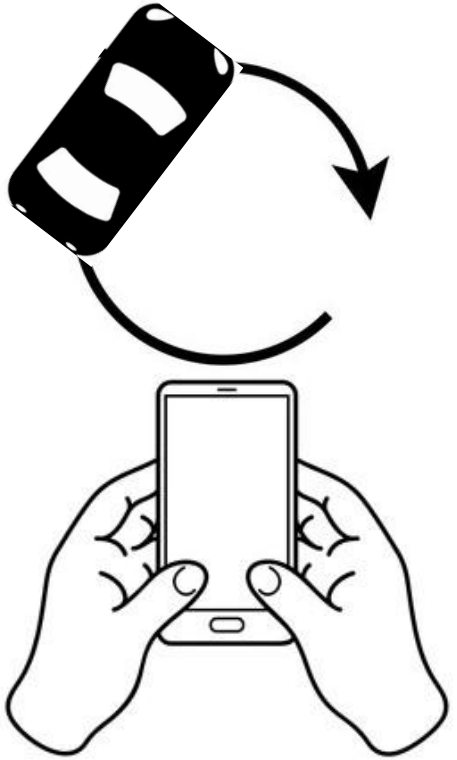
    ljmp MAIN
```

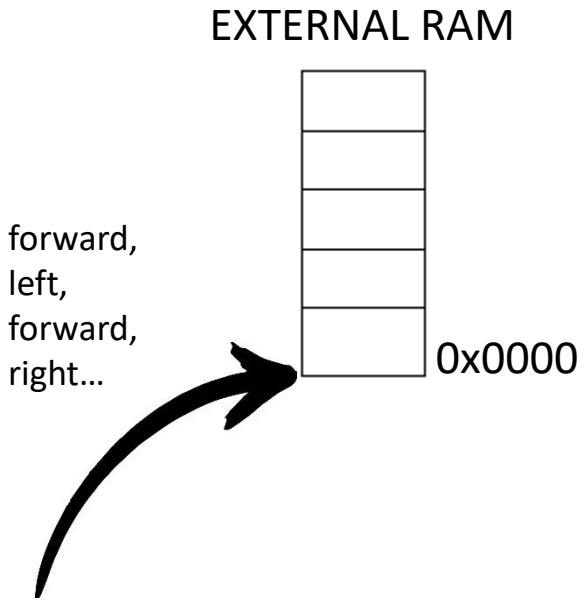


Function 2 : Path memorizing (experimental)(1/2)

Store “FORWARD”, “BACKWARD”,“LEFT”, “RIGHT” to external RAM before pass to L298N.

```
pop DPH
pop DPL
movx @DPTR, A ;STORE IN EXTERNAL RAM
inc DPTR
push DPL
push DPH
```

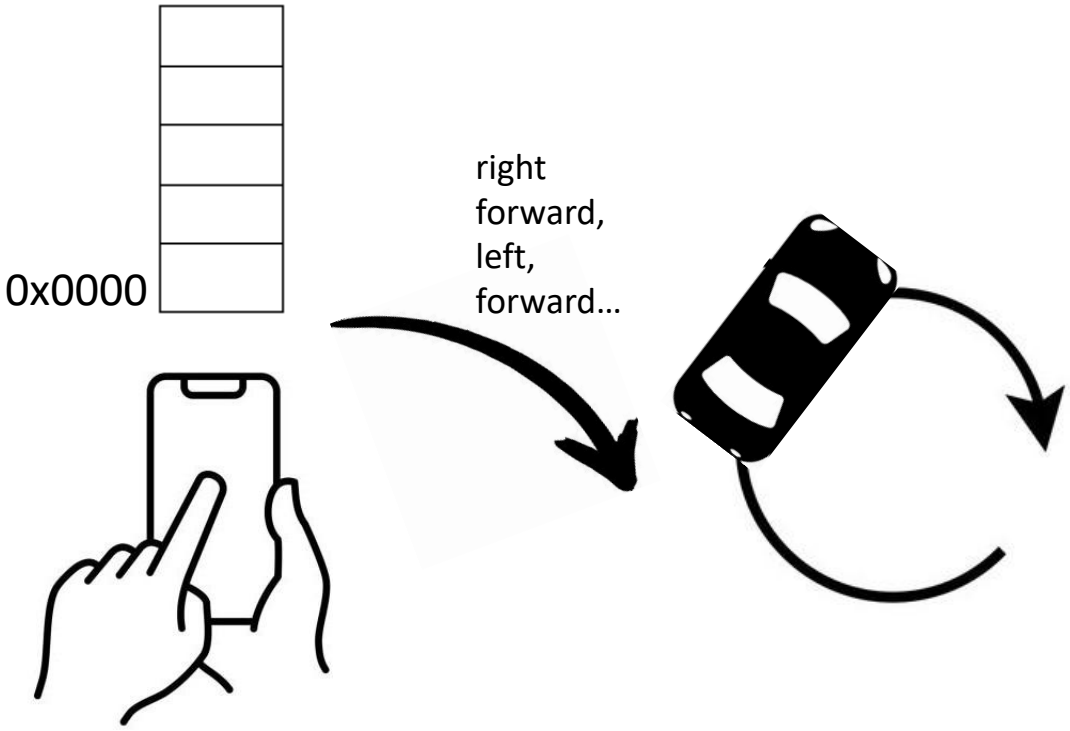




Control the car to go around a circle

Function 2 : Path memorizing (experimental)(2/2)

EXTERNAL RAM

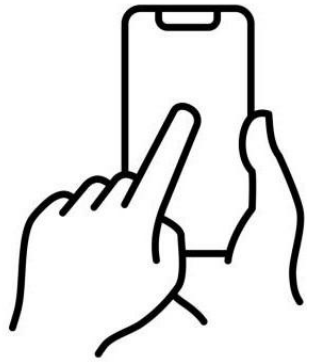


Tap “Replay” button to replay

Tap to trigger serial interrupt,
PC jump to 0x23,
turn off replay.

Interrupt handler

```
.org    0x0023 ;INTERRUPT VECTOR ADDRESS FOR "SERIAL"
    setb 0x01 ;INDICATE TO TURN OFF REPLAY
    clr ri
    reti ;BACK TO MAIN PROGRAM
```

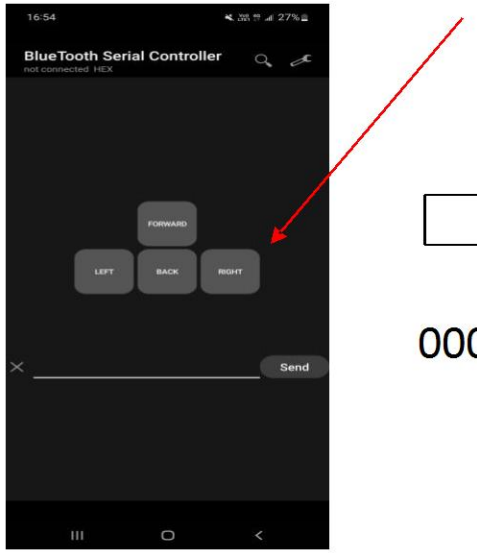


Tap any button to stop the replay

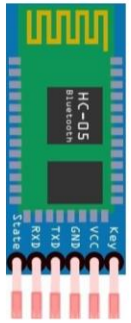
Summary

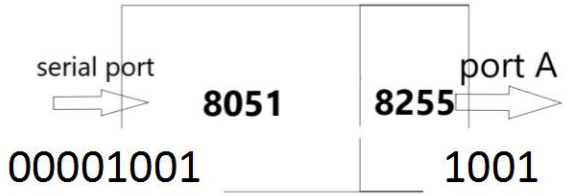
Ex. Turn right

Press this button



00001001

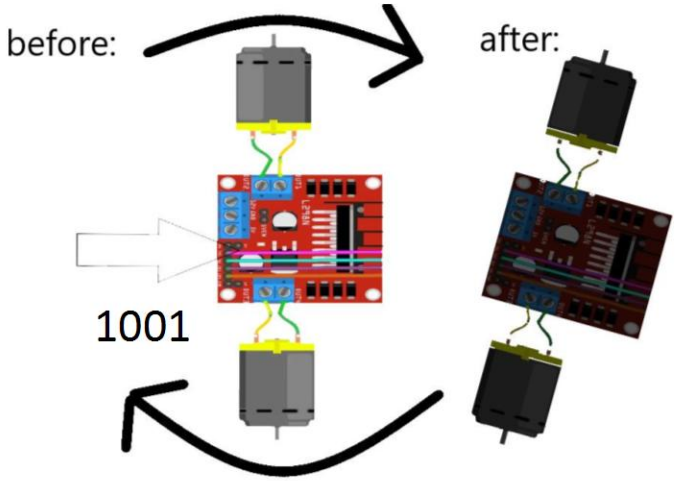





```
mov A, SBUF

    ;SPEED CONTROL
jb acc.4, SPEED     ;0001 000X

    ;ACTION MEMORIZING
jb acc.5, PLAY_LOOP ;0010 0000
jb acc.6, STOP      ;0100 0000
```



end